Ellipse: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Recognition and definition

Formulas

$$\Delta \neq 0,\ h^2 < ab$$

$$PS_1 + PS_2 = 2a > S_1S_2$$

$$PS = e \cdot PM,\ 0 < e < 1$$

KEY POINTS

- Circle is the special case $a = b,\ h = 0$
- $2a = S_1S_2$ gives only the segment S_1S_2
- Variable circle touching C_1 internally and C_2 externally: centre on an ellipse, foci C_1, C_2, $2a = r_1 + r_2$

Watch out

Check $\Delta \neq 0$ before calling it an ellipse

2. Standard ellipse

Formulas

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad b^2 = a^2(1 - e^2)$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

KEY POINTS

- Vertices $(\pm a, 0)$, minor ends $(0, \pm b)$, foci $(\pm ae, 0)$, directrices $x = \pm\frac{a}{e}$
- LR ends $\left(\pm ae, \pm\frac{b^2}{a}\right)$, $\ell_{LR} = \frac{2b^2}{a} = 2e\times$(focus–directrix distance)
- Vertical ($a > b$ under y^2): swap the roles of x and y
- Similar ellipses $\Leftrightarrow$ same e; larger e = flatter

Watch out

a is always the semi-MAJOR axis (bigger denominator)

3. Focal distances

Formulas

$$PS_1 = a - ex_1, \quad PS_2 = a + ex_1$$

$$PS_1 + PS_2 = 2a$$

KEY POINTS

- $PS_1 \cdot PS_2 = a^2 - e^2x_1^2 \in [b^2, a^2]$
- Triangle PS_1S_2: perimeter $2a + 2ae$; if $\angle P = 90°$, area b^2

4. Auxiliary circle and eccentric angle

Formulas

$$x^2 + y^2 = a^2$$

$$P(a\cos\theta,\ b\sin\theta), \quad Q(a\cos\theta,\ a\sin\theta)$$

KEY POINTS

- θ is the angle of CQ (not of CP); $\frac{PN}{QN} = \frac{b}{a}$
- Point $(p\cos\theta, q\sin\theta)$ moves on $\frac{x^2}{p^2} + \frac{y^2}{q^2} = 1$ (rod on axes, ordinate in fixed ratio)
- Max / min problems: use θ; $a\cos\theta + b\sin\theta \in [-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$

5. Shifted ellipse and other forms

Formulas

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

STEPS

1. No xy, x^2, y^2 coefficients same sign: complete squares
2. Put $X = x - h$, $Y = y - k$; read the standard results
3. Shift back to x, y

KEY POINTS

- $PS = e\,PM$ form: read focus, directrix, e; centre at ae from the focus away from the directrix
- $PS_1 + PS_2 = 2a$ form: centre = mid-point of foci, $2ae = S_1S_2$
- Alternate form $\frac{p_2^2}{a^2}+\frac{p_1^2}{b^2}=1$: p_2 = distance from the minor axis

Watch out

Divide each bracket by $\sqrt{l^2+m^2}$ before reading a^2, b^2
$K = 0$ after completing squares: a point, not an ellipse

6. Position of a point, chord joining two points

Formulas

$$S_1=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$$

$$\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$$

KEY POINTS

- $S_1 < 0$ inside, $= 0$ on, > 0 outside
- Chord through $(d, 0)$: $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{d-a}{d+a}$; focal chord: $\frac{e-1}{e+1}$

7. Line and ellipse, condition of tangency

Formulas

$$c^2=a^2m^2+b^2$$

$$y=mx\pm\sqrt{a^2m^2+b^2}$$

KEY POINTS

- $c^2 < a^2m^2 + b^2$ secant, $=$ tangent, $>$ no common point
- $lx + my + n = 0$: $a^2l^2 + b^2m^2 = n^2$; $x\cos\alpha + y\sin\alpha = p$: $p^2 = a^2\cos^2\alpha + b^2\sin^2\alpha$
- Contact point of $y = mx + c$: $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$
- Common tangent with another curve: slope form + the other curve's condition

Watch out

Vertical ellipse: $c^2 = b^2m^2 + a^2$
Vertical tangents $x = \pm a$ are missed by the slope form

8. Director circle, tangents from a point

Formulas

$$x^2+y^2=a^2+b^2$$

$$(h^2-a^2)m^2-2hkm+k^2-b^2=0$$

KEY POINTS

- $m_1 + m_2 = \frac{2hk}{h^2-a^2}$, $m_1m_2 = \frac{k^2-b^2}{h^2-a^2}$: put in the given condition for the locus

- Perpendicular tangents meet on the director circle; an ellipse sliding between two perpendicular lines has its centre on a circle

9. Tangent

Formulas

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

KEY POINTS

- Tangents at α, β meet at $\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$
- Intercepts $\frac{a}{\cos\theta}, \frac{b}{\sin\theta}$: least triangle area ab, least intercept $a+b$
- Parallel tangents touch at θ and $\theta+\pi$
- Foot of perpendicular from the centre on a tangent: $(x^2+y^2)^2 = a^2x^2 + b^2y^2$

10. Normal

Formulas

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$$

$$ax\sec\theta - by\,\mathrm{cosec}\,\theta = a^2 - b^2$$

KEY POINTS

- Normal condition for $y = mx + c$: $c^2 = \frac{m^2(a^2-b^2)^2}{a^2+b^2m^2}$
- Meets the major axis at $G(ae^2\cos\theta, 0)$; at most 4 normals from a point
- Shortest distance from a point to the ellipse is along a normal

Watch out

Normal at θ uses $\sec\theta$ and $\mathrm{cosec}\,\theta$: not defined at the vertices; take the axes there

11. Minimum distance

STEPS

1. Line and ellipse: check that they do not meet
2. Tangent parallel to the line (near side)
3. Distance between the parallel lines; nearest point = contact point

KEY POINTS

- Point and ellipse: minimise $(a\cos\theta - h)^2 + (b\sin\theta - k)^2$ or use the normal through the point

12. Chords: T = S1, T = 0, SS1 = T²

Formulas

$$T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$$

$$T = S_1, \quad T = 0, \quad SS_1 = T^2$$

KEY POINTS

- $T = S_1$: chord with mid-point (x_1, y_1); loci of mid-points: put the condition in it
- $T = 0$: tangent (on), chord of contact (outside), polar (any point)
- Pole of $lx + my + n = 0$: $\left(-\frac{a^2l}{n}, -\frac{b^2m}{n}\right)$; chords of contact from a fixed line pass through its pole
- Diameter for slope m: $y = -\frac{b^2}{a^2m}x$; conjugate diameters $m_1m_2 = -\frac{b^2}{a^2}$

Watch out

$T = S_1$ simplifies to $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$ (the -1 terms cancel); do not confuse it with $T = 0$

13. Highlights

KEY POINTS

- Product of perpendiculars from the foci on any tangent $= b^2$
- Feet of these perpendiculars lie on the auxiliary circle
- Tangent between point of contact and directrix subtends $90°$ at the focus
- Circle on a focal distance as diameter touches the auxiliary circle
- Reflection: normal bisects $\angle S_1PS_2$; a ray from one focus passes through the other

14. Common mistakes checklist

STEPS

1. Is a the bigger semi-axis? (vertical ellipse)
2. Foci at $\pm ae$, directrices at $\pm a/e$ - not mixed up
3. Eccentric angle is not the polar angle of P
4. Shifted ellipse: shift the line/point too before using standard results
5. From a point with $h = \pm a$ one tangent is vertical
6. Check $c^2 > a^2m^2 + b^2$ before using parallel-tangent distance